OPEN AGENT HACKATHON 2026 · TRACK 02

Kilowhy

Ask why your electricity bill changed.

Every number is backed by a query.



Real Octopus Agile prices, London, 17 Oct 2026 · 48 half-hours · peak 50.06p/kWh at 18:30

01 · THE PROBLEM

The bill went up. Nobody could say why.

+£339.91

in four weeks

3 London bakeries
half-hourly Agile tariff

The answer hides in five sources:

- **Prices**
- **Grid carbon**
- **Generation**
- **Weather**
- **Meters**

02 · THE AGENT

Ask one question. Get a verified answer.



03 · DEMO · LIVE

Watch it investigate

Every step shows the model that answered, via Meterless Nightshift.

- Understand question — 54.7 s
- Split the bill — 21.4 s
- Test hypotheses — 0.4 s
- Write answer — 0.0 s
- Verify every figure

Understand question Nightshift receipt — Nemotron 3 Ultra 550B A55B · 19.3 s · 836 tok · race of 6 ▸

Plan checks NVIDIA direct · fallback — NVIDIA direct (Meterless fallback) · 2.23 s · 572 tok ▸

Interpret 1/2 (H1, H2, H3, H5, H6) Nightshift receipt — Nemotron 3 Ultra 550B A55B · 28.5 s · 4,650 tok · race of 6 ▸

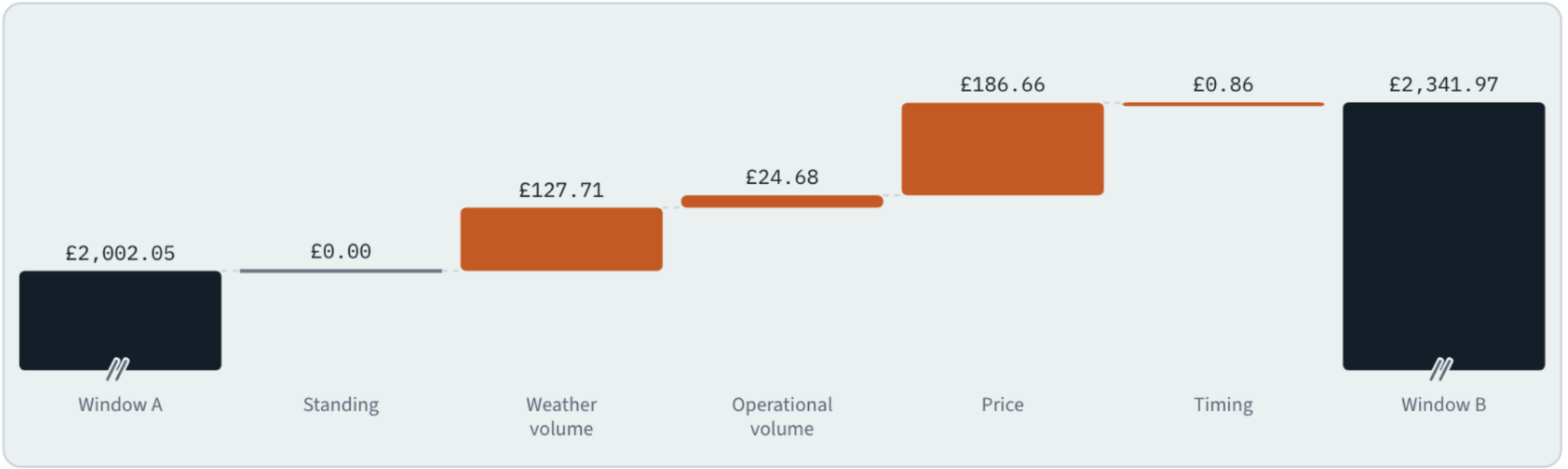
+£339.91, split exactly

+£186.66	+£127.71	+£24.68	+£0.86
price	weather	operations	timing

The bill rose by £339.91 across the three sites

Verified: every figure matches its source

How the bill moved (window A → window B)



Meter readings are sample data; prices and weather are real

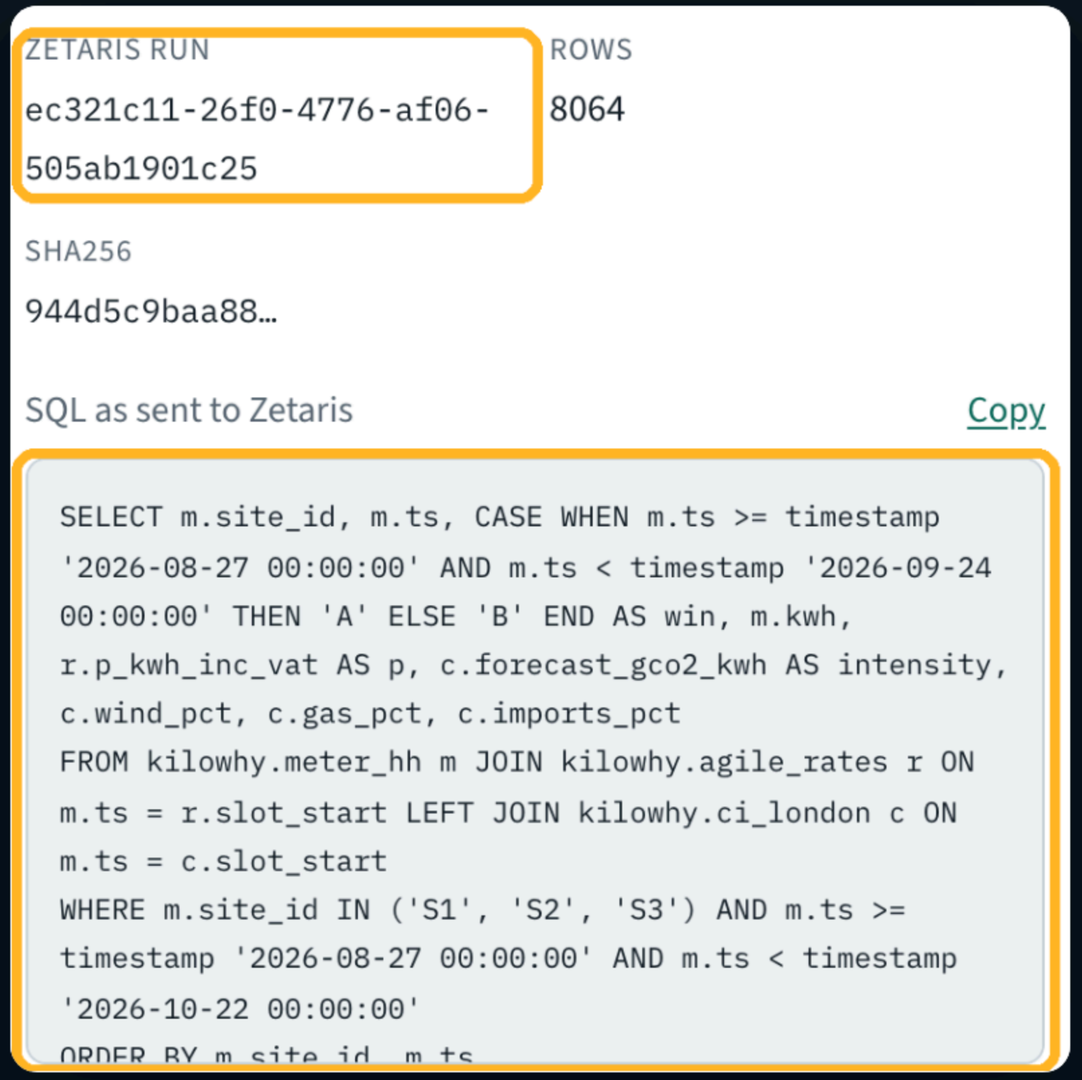
05 · DEMO · EVIDENCE

One click from sentence to SQL

EVIDENCE CHIP

E3 Price level A vs B

→ Each chip opens the exact query Zetaris ran: run id, rows, SHA-256.



£504.69

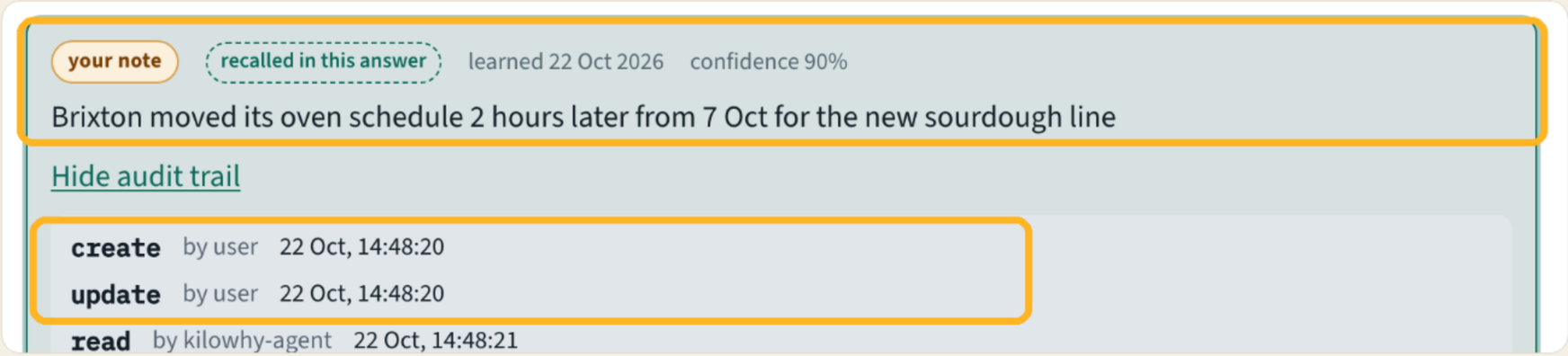
saved by staying on Agile instead of a flat business tariff.

Moving flexible load would save just £5.23. Kilowhy says so.

It remembers what the owner tells it

1

A note, kept with its audit trail

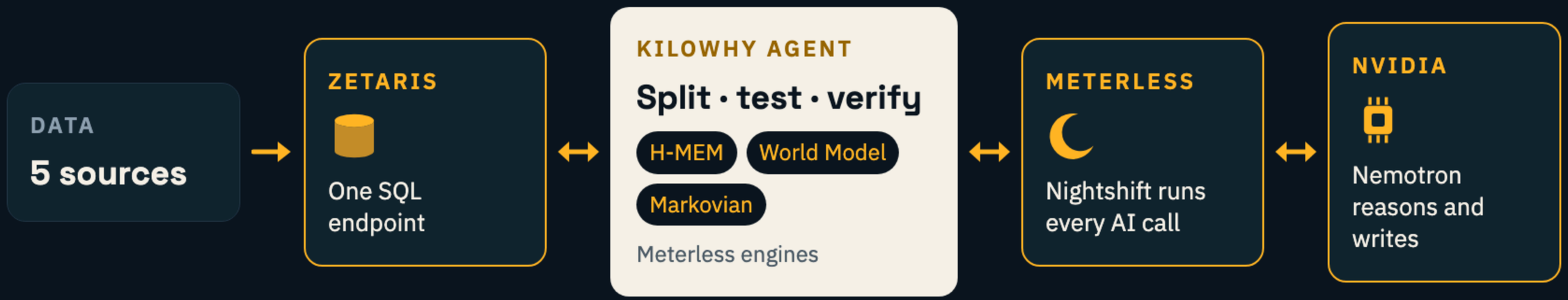


2

The next Brixton answer recalls it



Four sponsors, one pipeline



CURSOR built the live web UI

09 · ZETARIS

Every number is a Zetaris query

5

sources, 4 of them live APIs

1

SQL endpoint for the agent

0

rows copied; data stays at source

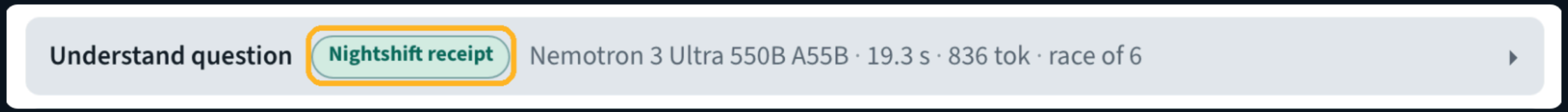
0p

gap vs raw Octopus API, 6 of 6 checks

```
SELECT … FROM kilowhy.meter_hh m JOIN kilowhy.agile_rates r ON m.ts = r.slot_start
```

10 · METERLESS

Nightshift runs the AI. Three engines do real work.



Every model call comes back with a Nightshift receipt.

H-MEM

Remembers

Owner notes, with an audit trail

WORLD MODEL

Grounds

Sites, meters and logged events, read from Zetaris

MARKOVIAN

44% fewer tokens

Bounded steps, compact carryover (estimated)

Nemotron never writes a number.

1 · NEMOTRON WRITES

"higher prices added {F13}"

→

2 · CODE FILLS

F13 = £186.66 · E1

→

3 · VERIFIER CHECKS

✓ Traced to evidence

Nemotron 3 Ultra 550B

understands, plans, interprets, writes

Nemotron Embed 1B

finds the relevant owner notes

Tested on 12 planted-cause scenarios

100%

precision

30 of 30 reported causes were real

93.8%

recall

30 of 32 real causes found

96.8%

F1

12/12

exact splits

parts add up to the penny

Expected verdicts written before the first run · real prices and weather

Built with Cursor and Claude Code

CURSOR AGENT

The web UI

Live timeline, answer, evidence drawer, memory panel

Prompts and transcripts are in the repo

CLAUDE CODE

The agent

Data layer, 11 tests, verifier, engines, evaluation

Plus the final UI polish

671 automated tests

1 command starts the stack

KILOWHY · ASK WHY, GET THE EVIDENCE

Every half-hourly bill, explained.

Next: live smart-meter feeds, more tariffs, more sites.

Try it: yy5652-hash.github.io/kilowhy · Code: github.com/yy5652-hash/kilowhy



Zetaris · Meterless · NVIDIA Nemotron · Cursor